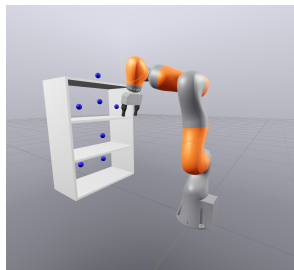
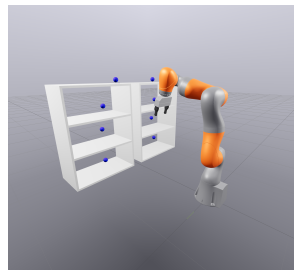


Результаты экспериментов по планированию с GCS

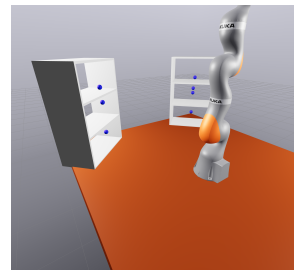
1 Начальные точки в полках



(a) Одна полка



(b) Две полки



(c) Три полки

Figure 1: Примеры начальных точек для построения IRIS-регионов в сценах с полками

2 Результаты экспериментов

Table 1: GCS planning statistics by scene and number of seeds (regions). Failure breakdown is shown as counts of IRIS-region construction failures vs disconnected-region failures (No path). Abbreviations: SS — Single Shelf, TS — Two Shelves, 3S — Three Shelves.

Scene	Seeds	Success	IRIS fail	No path	Path Length	Solve Time (s)
SS	10	4/21	0	17	0.28 ± 0.08	0.003
SS	20	4/21	0	17	0.28 ± 0.08	0.003
SS	30	4/21	0	17	0.28 ± 0.08	0.003
SS	50	11/21	0	10	2.08 ± 2.20	0.019
TS	10	0/28	13	15	—	—
TS	20	0/28	13	15	—	—
TS	30	0/28	13	15	—	—
TS	50	0/28	13	15	—	—
3S	10	0/55	19	36	—	—
3S	20	1/55	19	35	6.99 ± 0.00	0.100
3S	30	0/55	19	36	—	—
3S	50	3/55	19	33	7.04 ± 1.55	0.080

Table 2: Summary of RRT* $\rightarrow$ GCS experiments (pairs where both methods succeeded). Abbreviations: SS — Single Shelf, TS — Two Shelves, 3S — Three Shelves.

Scene	Pairs	RRT* OK	GCS OK	RRT* length	GCS length	Improvement
SS	21	21/21	16/21	10.53 ± 8.14	4.52 ± 3.43	$40.8\% \pm 30.2\%$
TS	28	28/28	16/28	9.24 ± 3.26	5.48 ± 2.84	$40.7\% \pm 28.1\%$
3S	32	32/32	14/32	10.85 ± 4.66	5.38 ± 1.69	$45.9\% \pm 16.1\%$
TOTAL	81	81/81	46/81	10.18 ± 5.69	5.11 ± 2.76	$42.3\% \pm 25.5\%$

Table 3: Runtime breakdown (mean, seconds)

Scene	RRT* time	IRIS time	GCS time	Total
SS	3.50	68.0	0.109	71.6
TS	0.14	156.3	0.153	156.6
3S	2.63	214.3	0.201	217.2